



Figure 1: Raspberry Pi

programmers to develop all sorts of Python based programs.

In addition to Python, various other languages are supported by Raspberry Pi. A number of IDEs (integrated development environments) that are free and open source are also available. These allow programmers, developers and application engineers to develop programs and applications on Pi.

Top IDEs for Raspberry Pi

As a programmer and developer, the first thing you require is an IDE, which is regarded as a comprehensive software suite that integrates the basic tools that developers and programmers require to write, compile and test their software. An IDE contains a code editor, a compiler or interpreter and a debugger, which the developer can access via a graphical user interface (GUI). One of the main aims of an IDE is to reduce the configuration necessary to piece together multiple development utilities, and provide the same set of capabilities as a cohesive unit.

An IDE's user interface is similar to that of a word processor, for which tools in the toolbar support colour-coding, formatting of source code, error diagnostics, reporting and intelligent code completion. IDEs are designed to integrate with third-party version control libraries like GitHub or Apache Subversion. Some IDEs are dedicated to a particular programming language, allowing a feature set that matches the programming language, while some support multiple languages.

Raspberry Pi has a wide range of IDEs that provide programmers with good interfaces to develop source code, applications and system programs.

Let's explore the top IDEs for Raspberry Pi.

BlueJ

BlueJ is an IDE that is dedicated to the Java Programming Language and was mainly developed for educational purposes. It also supports short software development projects. Michael Kolling and John Rosenberg at Monash University, Australia, started BlueJ development in 2000 as a powerful successor to the Blue system, and BlueJ became free and open source in March 2009.

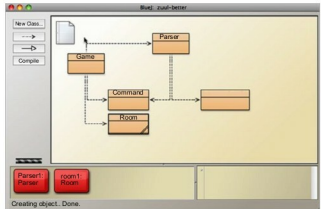


Figure 2: The BlueJ GUI interface

BlueJ provides an efficient way for learning object-oriented programming concepts and the GUI provides a class structure for applications like UML diagram. Every OOPS based concept, like class, objects and function calling, can be represented via interaction based design.

Features

- **Simple and interactive interface:** The user interface is simple and easy to learn as compared to other professional interfaces like NetBeans or Eclipse. Developers can focus mainly on programming rather than the environment.
- **Portable:** BlueJ supports multiple platforms like Windows, Linux and Mac OS X, and can even run without any installation.
- **New innovations:** BlueJ IDE is filled with innovations in terms of the object bench, code pad and scope colouring, which makes development fun even for newbies.
- **Strong technical support:** BlueJ has a hard-core functioning team that responds to queries and offers solutions to all sorts of developer problems within 24 hours.

Latest version: 4.0.1

Official website: <https://www.bluej.org/>

Geany IDE

Geany IDE is regarded as a very lightweight GUI based text editor that uses Scintilla and GTK+ with IDE environment support. The unique thing about Geany is that it is designed to be independent of a special desktop environment and requires only a few dependencies on other packages. It only requires GTK2 runtime libraries for execution. Geany IDE supports tons of programming languages like C, C++, C#, Java, HTML, PHP, Python, Perl, Ruby, Erlang and even LaTeX.

Features

- Auto-completion of code and simple code navigation.
- Efficient syntax highlighting and code folding.
- Supports embedded terminal emulator, and is highly extensible and feature-rich since lots of plugins are available for free download.